

Aadithya AR

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SUMMARY

Machine Learning Engineer skilled in NLP, LLMs, and Retrieval-Augmented Generation (RAG). Experienced in building scalable ML pipelines, deploying models via FastAPI and AWS, and delivering production-ready AI systems with measurable performance benchmarks.

EDUCATION

Global Academy of Technology

B.E. in Computer Science Engineering (AI & ML) | CGPA: 8.5 / 10.0

Bangalore, India

Oct. 2023 – Jul 2027

- **Coursework:** Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks, Machine Learning, Deep Learning

EXPERIENCE

Machine Learning / AI Intern (Agentic AI)

Feb 2026 – Present

Innomatics Research Labs

Remote

- Engineered production-grade ML data pipelines using LangChain and Python — prompt engineering, LLM chaining, and document-grounded QA; delivered 2 end-to-end pipelines with sub-3s inference latency, improving response efficiency for real-time applications.
- Built a high-performance FastAPI backend for ML model deployment with Pydantic validation and pytest coverage; deployed on AWS EC2 with environment-based config, achieving p99 response under 180ms.
- Designed a reusable NLP preprocessing library (tokenization, stemming, lemmatization, TF-IDF vectorization) adopted across 3+ downstream ML tasks — compatible with HuggingFace and Scikit-learn workflows.

PROJECTS

Adaptive RAG System with XAI + MLOps

Sep – Nov 2024

Python, Mistral-7B via Ollama, ChromaDB, BGE Embeddings, LangChain, RAGAS, W&B, Gradio

[GitHub](#)

- Built a high-performance document QA system over 50 ArXiv ML papers with adaptive query routing — switches between BM25 and dense retrieval (70/30 hybrid); 1,206 chunks embedded with model deployment via Ollama inference layer.
- Achieved RAGAS Context Precision 1.000 and Answer Relevancy 0.734; Explainable AI (XAI) using SHAP for source attribution and W&B MLOps tracking — mirrors CX knowledge-base retrieval and ticket resolution pipelines.

Parkinson's Voice Detection — Clinical ML Rebuild

Jan – Apr 2025

Python, PyTorch, wav2vec2-XLS-R-300M, praat-parselmouth, Optuna, XGBoost, LightGBM, Flask

[GitHub](#)

- Rebuilt ML data pipeline and feature extractor from scratch — fixed 7 silent failures (aliased MDVP calls, NoneType hardcodes, subject-level data leakage); extended feature set from 22 to 56 dimensions (CPP, MFCCs, formants).
- Deployed scalable model serving pipeline using wav2vec2-XLS-R-300M + Logistic Regression: CV AUC 0.972 ± 0.034 , Subject AUC 0.996, 94.25% accuracy on 831 recordings / 61 subjects; 75 Optuna hyperparameter tuning trials.

Nexus — Multi-Document RAG Workspace

Nov 2024 – Feb 2025

Next.js, Express.js, MongoDB Atlas, Groq (LLaMA-3.3-70B), HuggingFace BGE Embeddings

[GitHub](#)

- Built a production-grade multi-document QA system with cross-document contradiction detection, summarization, and MCQ generation; decoupled microservice backend (Express + MongoDB Atlas) with stateless Next.js frontend at \$0/month infra cost.
- Implemented data pipelines with paragraph-aware chunking and cosine top-K retrieval; Groq LPU inference (LLaMA-3.3-70B) delivers 3–10× lower latency versus OpenRouter for concurrent multi-user sessions.

TECHNICAL SKILLS

Languages: Python (primary), Java, C++, SQL, JavaScript

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems

ML / NLP: PyTorch, Scikit-learn, NumPy, Pandas, HuggingFace (wav2vec2, BGE), Optuna, XGBoost, LightGBM, RAGAS, W&B, Explainable AI (XAI), Feature Engineering, Model Deployment

GenAI & RAG: LangChain LCEL, Prompt Engineering, RAG, BM25, ChromaDB, Ollama, LangSmith

Backend & Cloud: FastAPI, Flask, RESTful APIs, Pydantic, pytest, AWS (EC2, S3), Docker, Git, CI/CD

CERTIFICATIONS & ACHIEVEMENTS

2nd Place — GAT Departmental Project Competition (15 teams, 60 students)

Nov 2025

IBM Datathon Participant — National Level, 2 consecutive editions

Jan 2026

NLP: Text Processing, Tokenization & Language Models — Infosys Springboard

Jun 2025

Machine Learning Foundation — Infosys Springboard

Dec 2025